



Morse code

Abstract

In this assignment, you will design and implement a simple program that converts a sequence of numbers into Morse code. The main objective is to understand how to translate digital input into timed physical signals and to apply fundamental programming concepts to control hardware behavior. Through this exercise, you will gain experience with timing, signal representation, and serial communication between the computer and the microcontroller. Your implementation will convert a numeric input into Morse code and display it by blinking an LED according to the standard Morse code timing rules.

Description

You will begin by entering an input value through the Serial Terminal Monitor. For this exercise, use the number **3451** as your test input. Your program will then interpret this input as individual digits and convert each one into its corresponding Morse code representation. The LED on your microcontroller board will blink to display the Morse code for each number in sequence.

- **Input:** The sequence **3451** entered from the Serial Terminal Monitor.
- **Output:** The LED should blink in a pattern that accurately represents the Morse code equivalent of each number.

To ensure accurate timing and consistency, follow the Morse code timing conventions provided below. These rules define how long the LED should stay on or off to represent dots, dashes, and the spaces between signals.

1. A **dot** corresponds to a signal that lasts for **100 milliseconds**.
2. A **dash** corresponds to a signal that lasts for the duration of **three dots**.
3. The space between **elements** of the same character (for example, between a dot and a dash within one number) should last for **one dot**.
4. The space between two **characters** (for example, between two different numbers) should last for **three dots**.

Your implementation should strictly follow these timing intervals to accurately represent Morse code. Pay attention to the consistency of the delays and ensure that the transitions between on and off states of the LED are clearly distinguishable. You may test and refine your timing to make the blinking visually clear and easily recognizable.

To assist you in correctly mapping each number to its Morse code sequence, refer to the International Morse Code chart available at the following link: https://en.wikipedia.org/wiki/Morse_code#/media/File:International_Morse_Code.svg

By the end of this assignment, you should have a working Morse code converter that takes input from the Serial Terminal Monitor and visually represents it using an LED. This exercise will help you connect abstract programming logic to physical behavior, reinforcing your understanding of timing, sequencing, and signal encoding in embedded systems.

Delivery

There are two deliverables.

- Unlisted YouTube video link for the blinking LED.
- Upload the Arduino code.

Due Date

Th Feb 19th, 11:59 PM EST